



Biometric authentication in cars

LG has developed a system allowing owners to turn on their cars using facial recognition. <https://digit.in/jan22-33>



Sevenaire Saturn 360 speaker

Sevenaire has launched a three-legged tablet which doubles up as a speaker. <https://digit.in/jan22-34>

HDMI goes the USB way

HDMI 2.1a, 2.1 and 2.0 are all the same and not the same. Confused?

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emember the old times when USB speeds were easy to understand and the features were not 10

different ancillary standards? I sure do. USB went through several standard revisions over the years. First it was 1.0, followed by 1.1 and then came 2.0 which continues to be the most popular standard since its release in the year 2000. USB 3.0 brought a lot of improvements in power and transfer rates but it soon underwent multiple revisions. It started with a USB 3.0 SuperSpeed mode which then became USB 3.1. Yeah, the USB-IF (USB Implementers Forum i.e. those who manage the USB standard) just renamed USB 3.0 SuperSpeed as USB 3.1 Gen 1 and brought out a faster version called USB 3.1 Gen 2. And if that wasn't enough, they chucked it all and came out with USB 3.2 and started renaming everything again. And the cherry on top of the cake? There's now an added layer of designation. It's now USB 3.2 Gen 1x1 or Gen 2x1 or Gen 2x2. And now that USB4 has started to come out, I'm pretty certain they'll find even better ways to confuse people.

The thing about maintaining standards is that you get to charge every company that wants to sell a USB product, a certain amount, for

certification. You can only advertise or call your cable as USB after you've received the certification. I'm not saying that this hellish jumble of similarly-specced standards was deliberately set up to allow the USB standard maintainers to milk hardware manufacturers but it sure seems that way.



The folks running the HDMI standard now want to copy this confusing way of handling standards. HDMI released version 2.0 around the same time as the folks over at USB-IF (USB Implementers Forum) did. HDMI 2.0 came out with support for TMDS encoding allowing it to carry 4K at 60 Hz and 24bit colour depth video signals. HDMI 2.0a soon followed with support for HDR video and static metadata. This was followed by HDMI 2.0b with more support for HDR metadata including

for Hybrid Log-Gamma. All of these features might seem like Latin and Greek but bear with me.

Then HDMI 2.1 was launched in 2017 and added support for 4K 120 Hz and 8K 120 Hz. In fact, it can go up to 10K at 120 Hz. Support also includes HDR dynamic metadata. And there's also Variable Refresh

Rate (VRR) and Auto Low Latency Mode (ALLM). HDMI 2.1 can carry up to 48 Gbps of data thanks to a new standard called FRL or Fixed Rate Link.

Thus far, it seems as if each new revision to the standard

is bringing a decent upgrade in terms of the resolution and frame rates aside from other features. It remains very clear what each revision brings to the table. And now comes the irritating part.

When HDMI 2.1 was announced, HDMI 2.0 ceased to exist. SHOCKING! Yes? No? Confusing, very much so.

HDMI 2.0 is treated as a subset of HDMI 2.1 and is clubbed under HDMI 2.1. There are a lot of features that distinguish HDMI 2.1 from its